

## Final Review Report : Creating Graphics Programs from Images

Team Name: *Your team's name*

Team Members: *Roll numbers of team members*

### 1 Problem Statement

Provide a short introduction about the project with proper motivation and applications. Give a broad overview of the problem statement that has been tackled in the project.

### 2 Work Done Before Stage-1 Review

This section can be populated from the proper details about the work done before the stage-1 review. You are expected to include all essential details that explain the work done till the stage-1 review.

An example for the description and citation can be seen as follows.

The neural network to infer spec  $S$  from an input image  $I$  uses ideas from Neurally guided procedural models [5] and attend-infer-repeat [3] models. The network is trained on sampled specs  $S$  and target images  $I$  from randomly generated scenes. The network is trained by maximizing the parametrized likelihood  $P_{\theta}(S|I)$  of inferring a spec  $S$  given an image  $I$ , with the parameters  $\theta$  being the parameters associated with the neural network. The network structure is illustrated in Figure 1. For further robustness to correct the network errors, the network outputs are further processed by a sequential Monte Carlo sampling scheme [1].

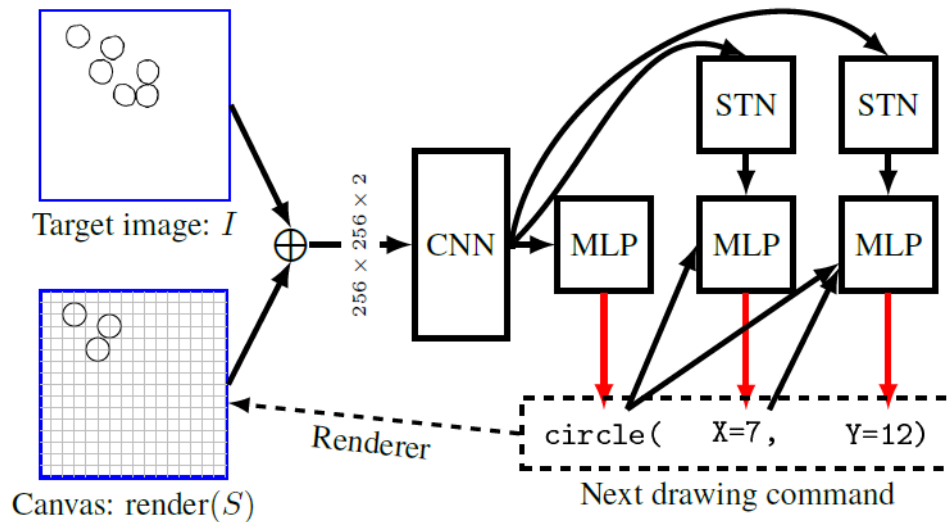


Figure 1: Neural architecture to infer specs from images.

### 3 Comments/Inputs given during the Stage-1 Review

Give details on the comments that were provided during the stage-1 review.

#### 3.1 Addressing Comments after Stage-1 Review

Brief summary on what measures have been taken by team to address the comments.

### 4 Experiments and Replications of Algorithms in Paper

In this section discuss all experimental settings, dataset details and experimental results are discussed with a proper summary. Include quantitative results in the form of tables and plots where necessary. Also include qualitative results wherever applicable.

### 5 Novel Settings and Experiments

Explain all details on the data sets used for the project. Description of data size and attributes, nature and type of data (image/audio/text/video etc.), data pre-processing techniques used should be illustrated. Other relevant details on data procurement (the website from where the data is obtained), and how the data is to be used in experiments should be described.

### 6 Conclusion

Conclude by giving a summary of your project, summarizing the problem, the methods used, and the significance of results obtained. Give some indications of future work to be done.

### 7 Contributions

Insert bullet points on individual contributions of the teammates towards the project.

### References

- [1] Arnaud Doucet, Nando de Freitas, and Neil Gordon (Eds.). *Sequential Monte Carlo Methods in Practice*. Springer-Verlag New York, 1 edition, 2001.
- [2] Kevin Ellis, Daniel Ritchie, Armando Solar-Lezama, and Joshua B. Tenenbaum. Learning to infer graphics programs from hand-drawn images. In *Advances in Neural Information Processing Systems*, 2018.
- [3] SM Eslami, N Heess, and T Weber. Attend, infer, repeat: Fast scene understanding with generative models. In *Advances in Neural Information Processing Systems*, 2016.

- [4] Armando Solar Lezama. *Program Synthesis By Sketching*. PhD thesis, EECS Department, University of California, Berkeley, 2008.
- [5] Daniel Ritchie, Anna Thomas, Pat Hanrahan, and Noah Goodman. Neurally-guided procedural models: Amortized inference for procedural graphics programs using neural networks. In *Advances in Neural Information Processing Systems*, 2016.